Project 3: Windows Failed Login Detection and SIEM Monitoring

Introduction

Windows Failed Login Detection and SIEM Monitoring is a hands-on SOC project that demonstrates how Windows Security Event Logs (Event ID 4625) can be collected, parsed, and analyzed using Linux (AWK) and Splunk Enterprise. In this project, failed login events were generated, processed into a CSV file, imported into Splunk, and investigated using SPL queries, dashboards, and basic incident analysis to identify suspicious authentication activity.

Lab Setup

The lab environment consists of an Ubuntu Linux machine, a Windows machine, and Splunk Enterprise. Failed login events were generated on Windows, processed using AWK on Ubuntu, and then imported into Splunk for log analysis and dashboard visualization

Pre-requisites

- Basic knowledge of Windows Security Event Logs
- Familiarity with Linux command-line (Bash & AWK)
- Splunk Enterprise installed and configured
- Windows and Ubuntu virtual machines
- Administrative access to the Windows system

Tools Used

- Splunk Enterprise
- Windows Event Viewer
- Ubuntu Linux
- AWK
- Bash
- Windows Security Event Logs
- CSV Files

Exrecise

Exercise 1: Generating Failed Login Events

Objective

Generate Windows failed login events (Event ID **4625**) to create security logs for investigation and analysis.

Step 1: Lock the Windows Machine

Lock the Windows system using the keyboard shortcut.

Windows + L

Explanation:
This locks the Windows machine and displays the login screen, allowing failed login attempts to be generated.

Step 2: Generate Failed Login Attempts

Enter an incorrect password multiple times on the login screen.

Explanation:
Each unsuccessful login attempt creates a **Windows Security Event ID 4625**, which records authentication failures.

Step 3: Verify Failed Login Events

Open **Event Viewer** and navigate to the Security log.

Windows Logs → Security

Filter the log using:

Event ID: 4625

Explanation:

- **Windows Logs** → **Security** contains authentication-related events.
- **Event ID 4625** represents a failed login attempt.
- Verifying this event confirms that the required log data has been generated successfully.

Expected Output

- Failed login events are successfully generated.
- Event ID **4625** appears in the Windows Security log.
- The logs are ready for further analysis.

Exercise 2: Verifying and Parsing Security Logs using AWK

Objective

Verify the exported Windows Security logs on Ubuntu and extract failed login events using AWK.

Step 1: Verify the Log File

Check that the exported security log file is available on the Ubuntu machine.

```
ls
```

Explanation:
Displays the available files and confirms that the log file is present.

Step 2: Display the Log File

View the contents of the exported CSV file.

```
head failed_logins_detailed.csv
```

Explanation:
Displays the first few records to verify that the log data has been exported correctly.

Step 3: Filter Events by Source IP

Extract events generated from the suspicious source IP.

```
awk -F',' 'NR>1 {gsub(/"/,"",$6); if($6=="192.168.0.102") print}'
failed_logins_detailed.csv
```

Explanation:
Filters the CSV file and displays only the events from 192.168.0.102.

Step 4: Identify Target Accounts

Display the usernames targeted by the failed login attempts.

```
awk -F',' 'NR>1 {gsub(/"/,"",$3); gsub(/"/,"",$6);
if($6=="192.168.0.102") print $3}' failed_logins_detailed.csv
```

Explanation:
Shows which user accounts were targeted by the suspicious source IP.

Expected Output

- Security logs are verified on Ubuntu.
- Failed login events are filtered successfully.
- Target user accounts are identified.

Exercise 3: Transferring and Importing Logs into Splunk Enterprise

Objective

Transfer the parsed CSV log file from the Ubuntu machine to the Windows machine and import it into Splunk Enterprise.

Step 1: Start the Python HTTP Server

Navigate to the directory containing the CSV file and start a Python HTTP server.

```
python3 -m http.server 8000
```

Explanation:
Starts a local web server on Ubuntu, making the CSV file accessible over the network.

Step 2: Download the CSV File

Open a web browser on the Windows machine and access:

```
http://<Ubuntu-IP>:8000
```

Download the failed_logins_detailed.csv file.

Explanation:
Downloads the CSV file from the Ubuntu machine to the Windows machine.

Step 3: Upload the CSV File to Splunk

Open Splunk Enterprise → Add Data → Upload, then select the downloaded failed_logins_detailed.csv file.

Explanation:
Imports the processed security log into Splunk Enterprise.

Step 4: Verify the Imported Data

```
index=main
```

Explanation:
Verifies that the imported events are successfully indexed and searchable.

Expected Output

- CSV file is transferred successfully.
- Log data is imported into Splunk Enterprise.
- Events are searchable using SPL queries.

Exercise 4: Analyzing Failed Login Events using SPL

Objective

Analyze the imported Windows Security logs to identify failed login attempts and suspicious authentication activity.

Step 1: View All Imported Events

Search all imported log data in Splunk.

```
index=main
```

Explanation:
Displays all events stored in the selected index.

Step 2: Filter Failed Login Events

Search only failed login events (Event ID 4625).

```
index=main EventID=4625
```

Explanation:
Filters the logs to display only Windows failed login events.

Step 3: Identify the Suspicious Source IP

Search for failed login attempts from the suspicious IP address.

index=main EventID=4625 Source_Network_Address="192.168.0.102"

Explanation:
Displays all failed login attempts originating from the specified source IP.

Step 4: Review Authentication Details

Analyze the event details such as Target User, Source IP, Logon Type, and Failure Status.

Explanation:
Helps identify which account was targeted and understand the authentication failure.

Expected Output

- Failed login events (Event ID 4625) are displayed.
- Suspicious source IP is identified.
- Authentication details are available for investigation.

Exercise 5: Dashboard Creation and Visualization

Objective

Create a Splunk dashboard to visualize failed login activity and simplify security monitoring.

Step 1: Create a Failed Login Statistics Panel

Use the following SPL query to display the total number of failed login attempts by user.

index=main EventID=4625

| stats count by TargetUserName

Explanation:
Displays the number of failed login attempts for each user account.

Step 2: Visualize Failed Login Activity

Create a chart to visualize failed login events by source IP.

index=main EventID=4625

| chart count by Source_Network_Address

Explanation:
Displays the distribution of failed login attempts based on the source IP address.

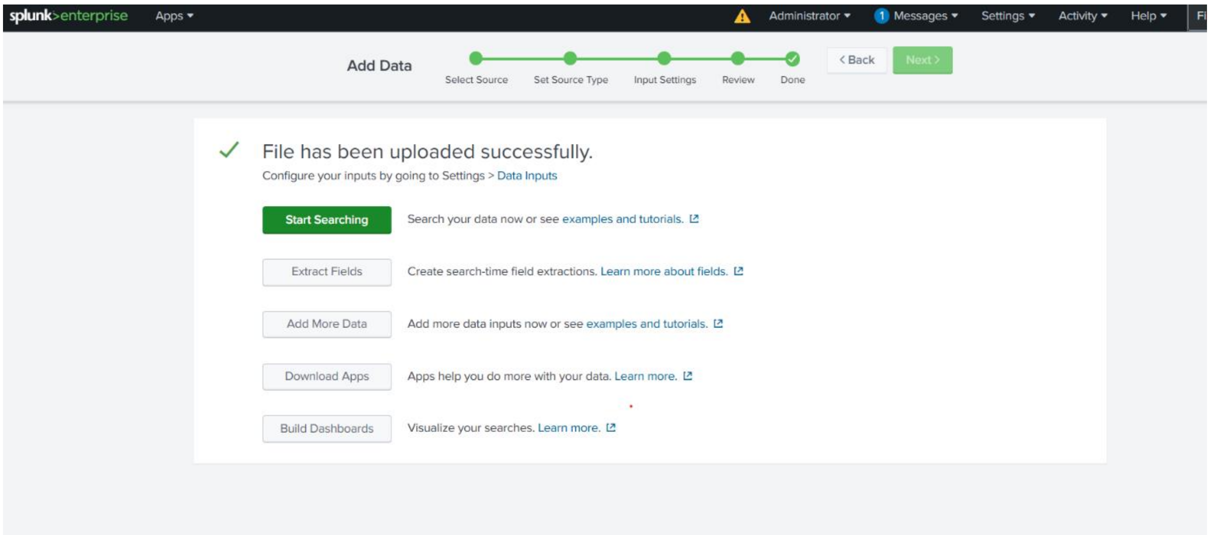
Step 3: Save the Dashboard

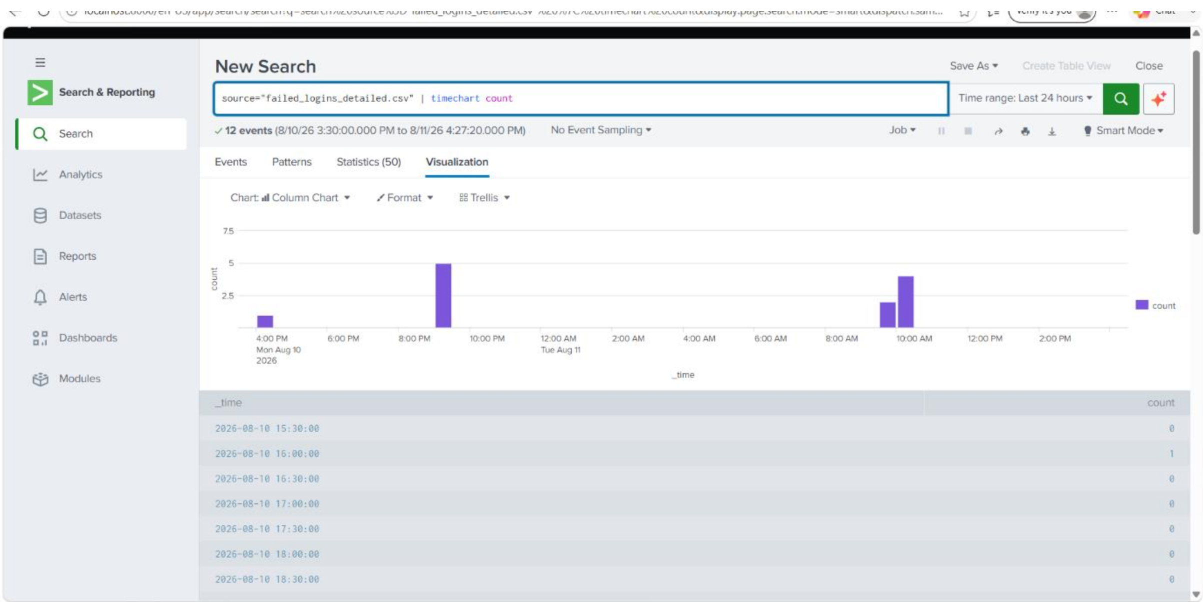
Save the visualizations to a new dashboard and organize the panels for easier monitoring.

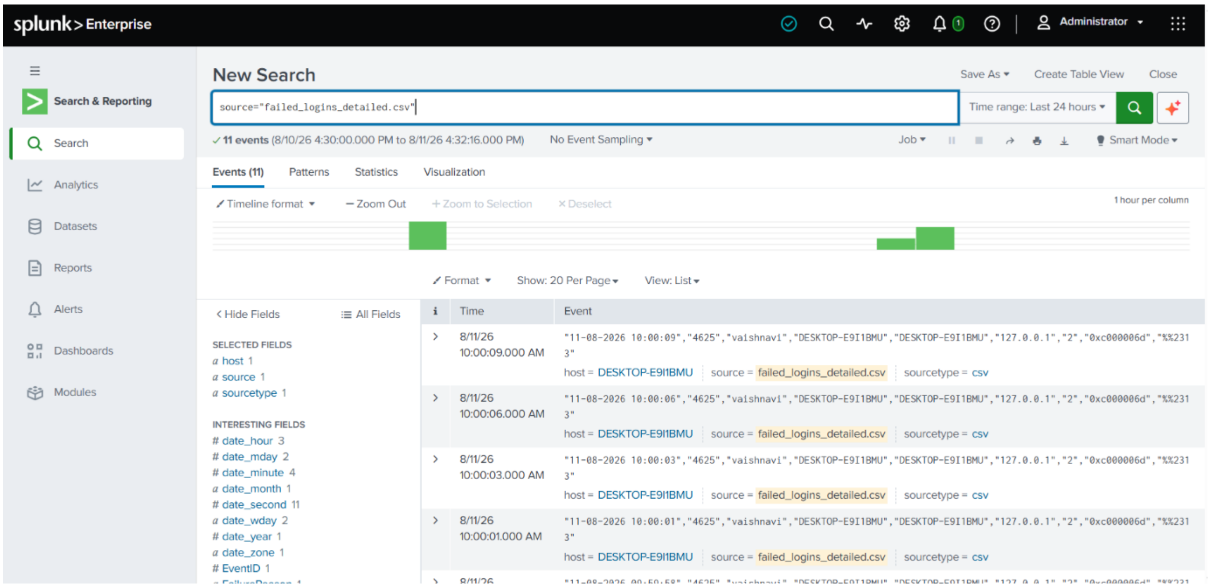
Explanation:
Provides a centralized view of failed login activity for quick analysis.

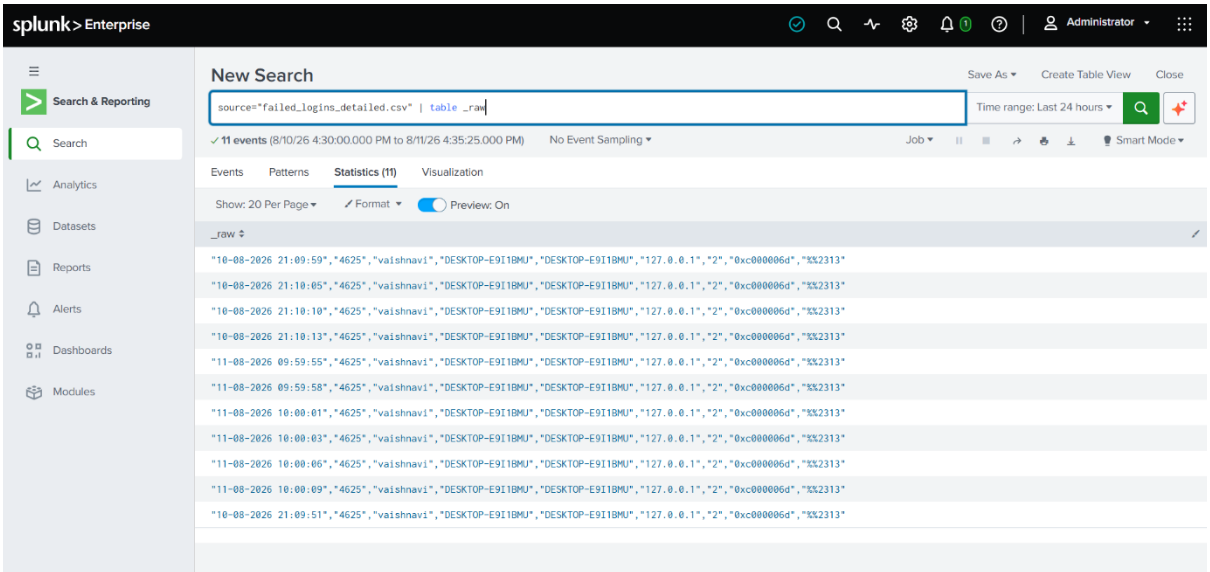
Expected Output

- Dashboard displays failed login statistics.
- Failed login events are grouped by user and source IP.
- Visualizations help identify suspicious authentication activity.


splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Add Data
Select Source
Set Source Type
Input Settings
Review
Done
Back
Next
File has been uploaded successfully.
Configure your inputs by going to Settings > Data Inputs
Start Searching
Search your data now or see examples and tutorials.
Extract Fields
Create search-time field extractions. Learn more about fields.
Add More Data
Add more data inputs now or see examples and tutorials.
Download Apps
Apps help you do more with your data. Learn more.
Build Dashboards
Visualize your searches. Learn more.


Search & Reporting
Search
Analytics
Datasets
Reports
Alerts
Dashboards
Modules
New Search
Save As
Create Table View
Close
source="failed_logins_detailed.csv" | timechart count
Time range: Last 24 hours
✓ 12 events (8/10/26 3:30:00.000 PM to 8/11/26 4:27:20.000 PM)
No Event Sampling
Job
Smart Mode
Events
Patterns
Statistics (50)
Visualization
Chart: Column Chart
Format
Trellis
7.5
5
2.5
count
4:00 PM Mon Aug 10 2026
6:00 PM
8:00 PM
10:00 PM
12:00 AM Tue Aug 11
2:00 AM
4:00 AM
6:00 AM
8:00 AM
10:00 AM
12:00 PM
2:00 PM
_time
count
_time
count
2026-08-10 15:30:00 0
2026-08-10 16:00:00 1
2026-08-10 16:30:00 0
2026-08-10 17:00:00 0
2026-08-10 17:30:00 0
2026-08-10 18:00:00 0
2026-08-10 18:30:00 0


splunk>Enterprise
Administrator
Search & Reporting
Search
Analytics
Datasets
Reports
Alerts
Dashboards
Modules
New Search
Save As
Create Table View
Close
source="failed_logins_detailed.csv"
Time range: Last 24 hours
✓ 11 events (8/10/26 4:30:00.000 PM to 8/11/26 4:32:16.000 PM)
No Event Sampling
Job
Smart Mode
Events (11)
Patterns
Statistics
Visualization
Timeline format
Zoom Out
+ Zoom to Selection
× Deselect
1 hour per column
Format
Show: 20 Per Page
View: List
Hide Fields
All Fields
SELECTED FIELDS
a host 1
a source 1
a sourcetype 1
INTERESTING FIELDS
date_hour 3
date_mday 2
date_minute 4
a date_month 1
date_second 11
a date_wday 2
date_year 1
a date_zone 1
EventID 1
i
Time
Event
8/11/26 10:00:09.000 AM
"11-08-2026 10:00:09","4625","vaishnavi","DESKTOP-E9I1BMU","DESKTOP-E9I1BMU","127.0.0.1","2","0xc000006d","%%2313"
host = DESKTOP-E9I1BMU source = failed_logins_detailed.csv sourcetype = csv
8/11/26 10:00:06.000 AM
"11-08-2026 10:00:06","4625","vaishnavi","DESKTOP-E9I1BMU","DESKTOP-E9I1BMU","127.0.0.1","2","0xc000006d","%%2313"
host = DESKTOP-E9I1BMU source = failed_logins_detailed.csv sourcetype = csv
8/11/26 10:00:03.000 AM
"11-08-2026 10:00:03","4625","vaishnavi","DESKTOP-E9I1BMU","DESKTOP-E9I1BMU","127.0.0.1","2","0xc000006d","%%2313"
host = DESKTOP-E9I1BMU source = failed_logins_detailed.csv sourcetype = csv
8/11/26 10:00:01.000 AM
"11-08-2026 10:00:01","4625","vaishnavi","DESKTOP-E9I1BMU","DESKTOP-E9I1BMU","127.0.0.1","2","0xc000006d","%%2313"
host = DESKTOP-E9I1BMU source = failed_logins_detailed.csv sourcetype = csv


splunk>Enterprise
Administrator
Search & Reporting
Search
Analytics
Datasets
Reports
Alerts
Dashboards
Modules
New Search
Save As
Create Table View
Close
source="failed_logins_detailed.csv" | table _raw
Time range: Last 24 hours
✓ 11 events (8/10/26 4:30:00.000 PM to 8/11/26 4:35:25.000 PM)
No Event Sampling
Job
Smart Mode
Events
Patterns
Statistics (11)
Visualization
Show: 20 Per Page
Format
Preview: On
_raw
"10-08-2026 21:09:59","4625","vaishnavi","DESKTOP-E9I1BMU","DESKTOP-E9I1BMU","127.0.0.1","2","0xc000006d","%%2313"
"10-08-2026 21:10:05","4625","vaishnavi","DESKTOP-E9I1BMU","DESKTOP-E9I1BMU","127.0.0.1","2","0xc000006d","%%2313"
"10-08-2026 21:10:10","4625","vaishnavi","DESKTOP-E9I1BMU","DESKTOP-E9I1BMU","127.0.0.1","2","0xc000006d","%%2313"
"10-08-2026 21:10:13","4625","vaishnavi","DESKTOP-E9I1BMU","DESKTOP-E9I1BMU","127.0.0.1","2","0xc000006d","%%2313"
"11-08-2026 09:59:55","4625","vaishnavi","DESKTOP-E9I1BMU","DESKTOP-E9I1BMU","127.0.0.1","2","0xc000006d","%%2313"
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"11-08-2026 10:00:01","4625","vaishnavi","DESKTOP-E9I1BMU","DESKTOP-E9I1BMU","127.0.0.1","2","0xc000006d","%%2313"
"11-08-2026 10:00:03","4625","vaishnavi","DESKTOP-E9I1BMU","DESKTOP-E9I1BMU","127.0.0.1","2","0xc000006d","%%2313"
"11-08-2026 10:00:06","4625","vaishnavi","DESKTOP-E9I1BMU","DESKTOP-E9I1BMU","127.0.0.1","2","0xc000006d","%%2313"
"11-08-2026 10:00:09","4625","vaishnavi","DESKTOP-E9I1BMU","DESKTOP-E9I1BMU","127.0.0.1","2","0xc000006d","%%2313"
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